

Problem Set 4

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Due: April 10, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in `.pdf` form.
- This problem set is due before 23:59 on Friday April 10, 2026. No late assignments will be accepted.

Question 1

We're interested in modeling the historical causes of child mortality. We have data from 26855 children born in Skellefteå, Sweden from 1850 to 1884. Using the "child" dataset in the `eha` library, fit a Cox Proportional Hazard model using mother's age and infant's gender as covariates. Present and interpret the output.

Table 1:

	<i>Dependent variable:</i>
	child_surv
sexfemale	−0.084*** (0.027)
socBranchfarming	−0.007 (0.092)
socBranchbusiness	0.335** (0.141)
socBranchworker	0.110 (0.094)
Observations	26,574
R ²	0.001
Max. Possible R ²	0.986
Log Likelihood	−56,498.610
Wald Test	33.180*** (df = 4)
LR Test	32.267*** (df = 4)
Score (Logrank) Test	33.279*** (df = 4)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Question 2

We want to estimate how the amount of damage caused by a disaster relates to (1) whether disaster relief is provided and (2) the amount of relief that is provided conditional on a donation occurring when a disaster hits a given country. Estimate a Heckman selection model using the `disasters_response` data on GitHub in which `binContribution` is the outcome for the selection equation, and `originalContributionMillionUSDLogged` is the outcome for the second equation. Use `occurrences + deathsEM + normalizedDamageEMLogged` as your input variables for both equations. Present and interpret the output.